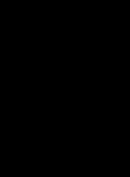


TURNING A SHELF OF EQUIPMENT MANUALS INTO A CONVERSATION

IntelliMaint AI is a virtual mechanic platform. A technician photographs a failing generator, asks a question out loud, and gets repair guidance drawn from the equipment's own documentation, instead of paging through a PDF binder at the side of the machine. We designed and built the product, from the interface to the retrieval workflow behind it.

AI PRODUCT BUILD PRODUCT DESIGN RAG RETRIEVAL VOICE & VISION



INTELLIMAINT AI
Virtual Mechanic Platform

Features How It Works Pricing Contact

Login Sign Up

Welcome to IntelliMaint AI

Your Troubleshooting Assistant

4

Capability pillars: documents, manual library, vision and voice
SHIPPED

3

Identity types: civilian, military and student
SHIPPED

4

Plan tiers, from free to enterprise
SHIPPED

9

Screens built, from sign-up to password reset
LIVE PRODUCT

CLIENT
IntelliMaint AI

SECTOR
Industrial & defence maintenance

ENGAGEMENT
AI product build

SURFACE
Responsive web application

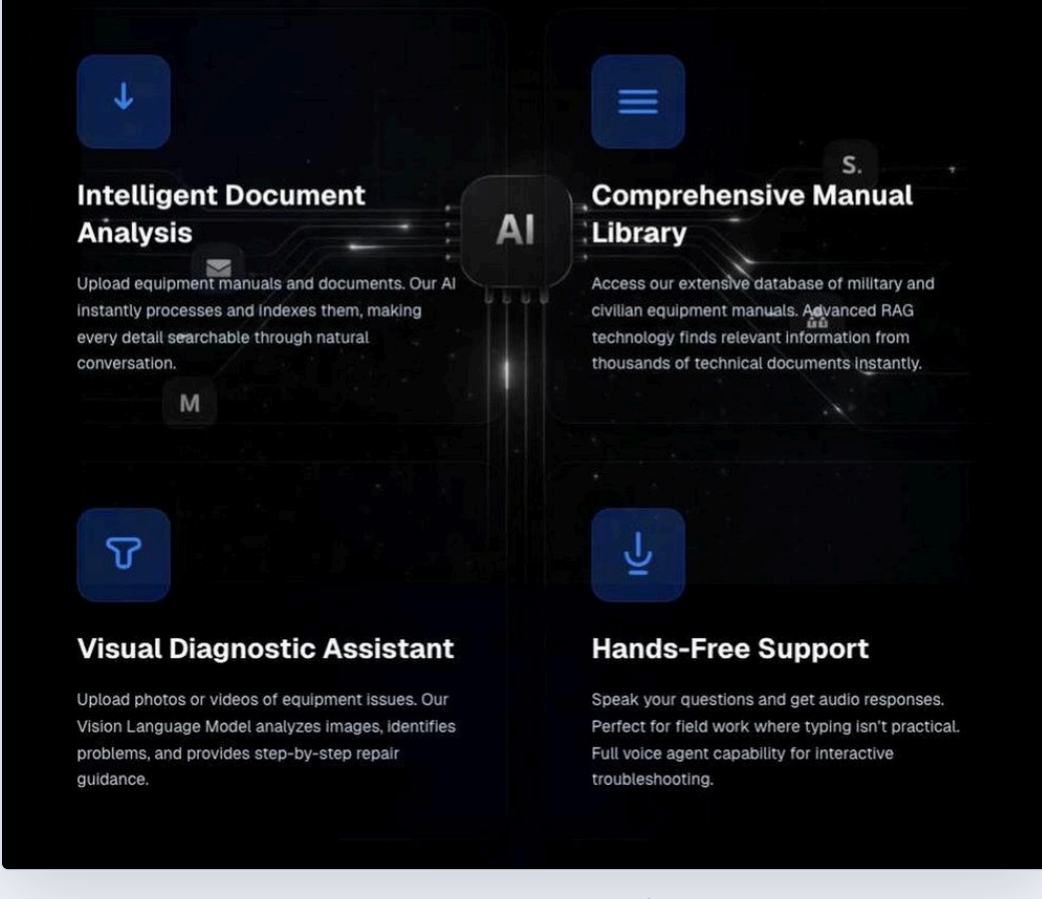
REVIEWED
September 2026

THE PRODUCT

AN ASSISTANT THAT HAS TO BE RIGHT, NOT JUST FLUENT

Most maintenance work does not fail for lack of intelligence. It fails because the one page that explains the fault is inside a 600-page PDF, on a laptop, in a truck, forty metres from the machine.

We built the platform around one behavioural change: replace searching for the manual with asking a question. A technician types, speaks or photographs the problem. The platform retrieves the relevant passages from that equipment's own documentation and returns step-by-step repair guidance, rather than generating an answer from general knowledge alone.



Four capabilities, each naming the mechanism and the field situation it serves.

THE CHALLENGE

THE KNOWLEDGE EXISTS. IT JUST ISN'T REACHABLE AT THE MACHINE

Industrial and defence maintenance runs on documentation: operator manuals, service bulletins, technical specifications and parts catalogues. The corpus is comprehensive and largely accurate. The problem is access, and it shows up in four recurring ways.

WRONG FORMAT A PDF is indexed by component. The technician's question arrives as a symptom: it starts, runs rough, then cuts out under load.	UNEVEN EXPERTISE The person who can diagnose a generator by ear is usually the most senior, and usually somewhere else.	HANDS ARE BUSY Field diagnosis is two-handed. A keyboard-first tool is abandoned at the machine.	RESTRICTED DOCUMENTS Defence documentation carries handling rules, so a general-purpose assistant is not an option.
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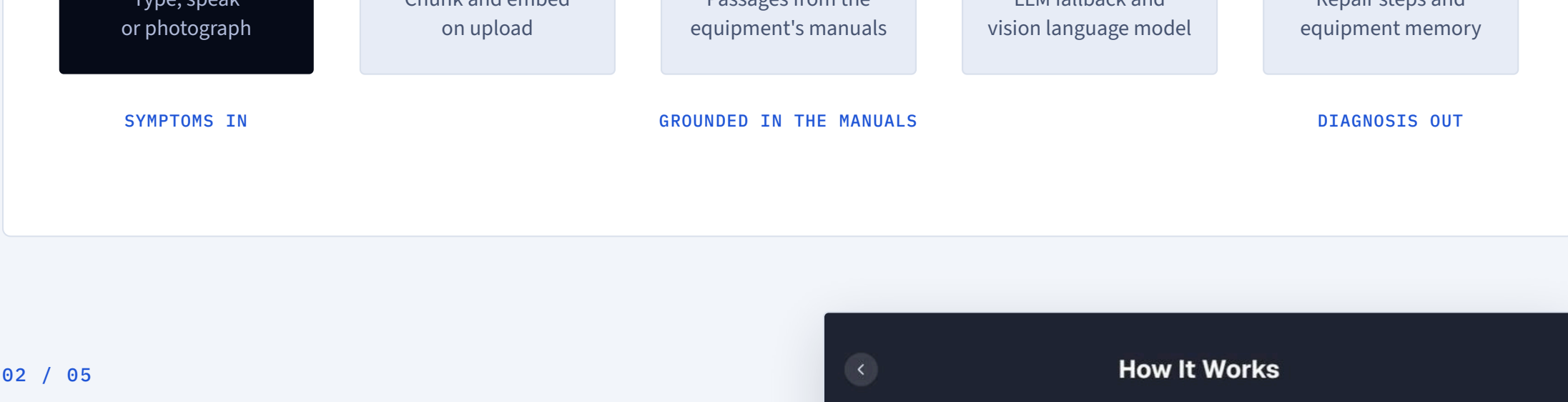
The design brief that follows: accept a question in whatever form the technician can produce it, answer it from that equipment's own documentation, and keep every account inside the repository it is permitted to see.

WHAT WE BUILT

01 / 05

RETRIEVAL FIRST, GENERATION SECOND

A chatbot that improvises a torque spec is a liability. So the workflow is organised around retrieval: documents are chunked, embedded and indexed on upload, the assistant answers from what it retrieved, and a large language model provides fallback rather than the primary source of truth.

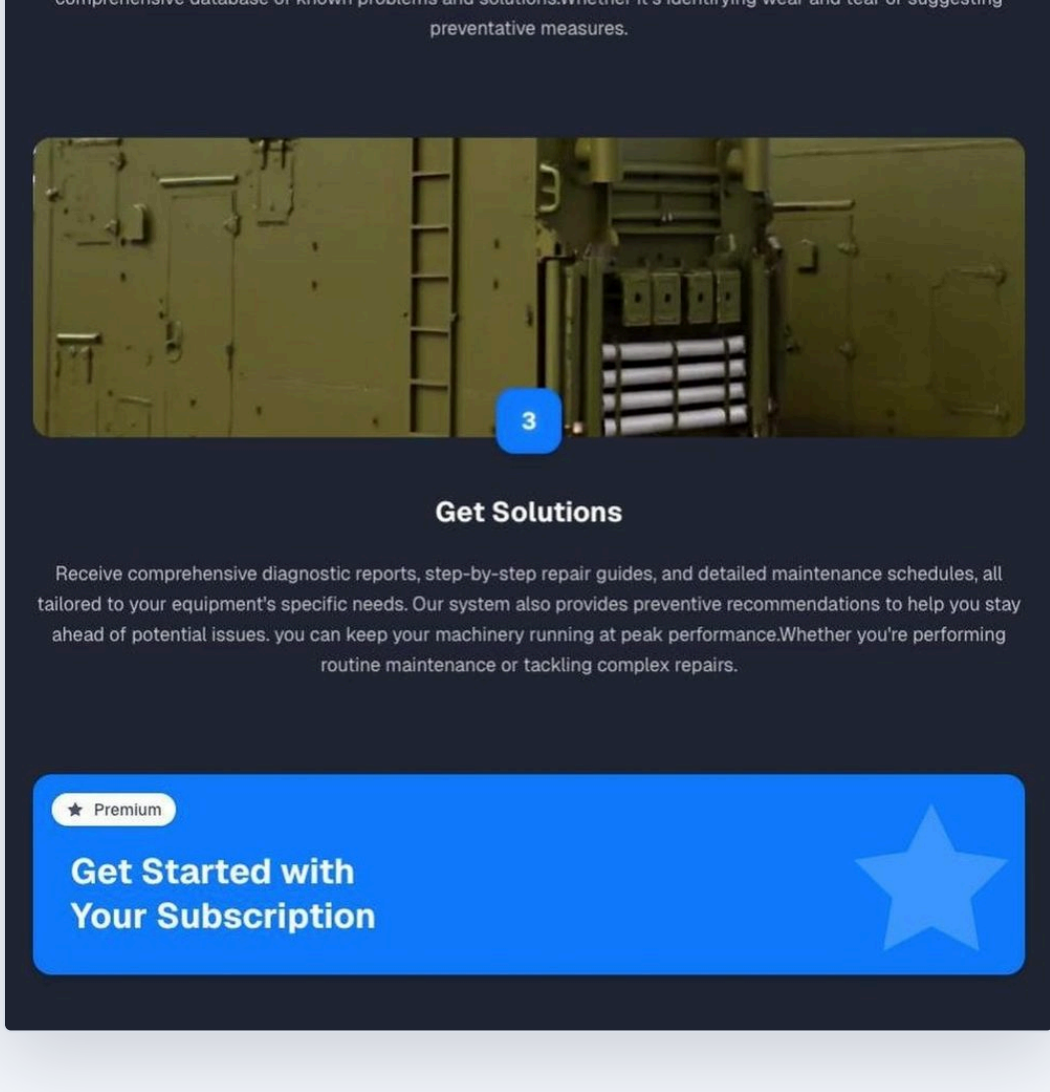
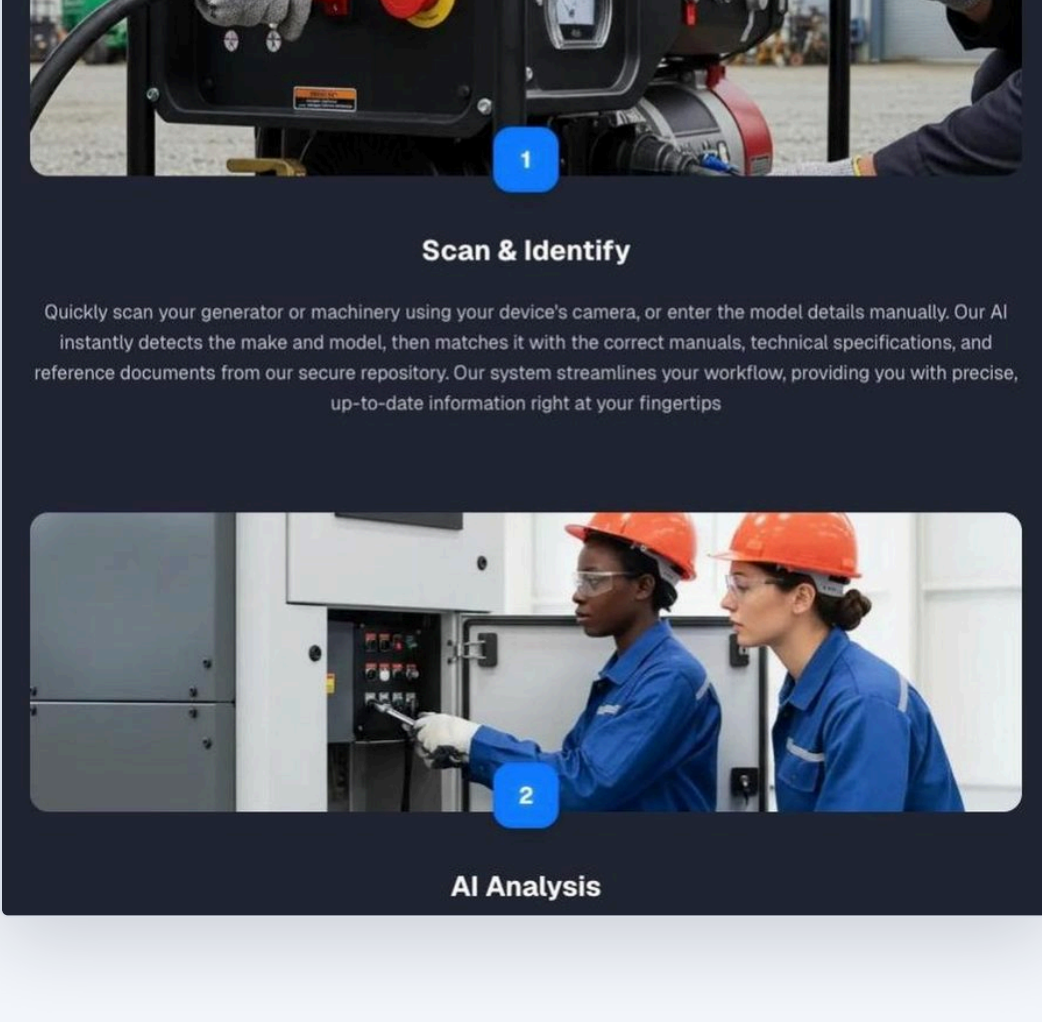


02 / 05

SYMPTOMS IN, DIAGNOSIS OUT, REPAIR STEPS ATTACHED

The product's organising idea is a route, not a feature set. Every screen either advances it or gets out of the way. The homepage leads with what a technician can do, and the identification step is a photograph, not a form. A technician who does not know the model number can still start the conversation.

SCAN AND IDENTIFY	Photograph the unit, or enter the model manually. The platform matches it to the right manuals and specifications.
ANALYSE	Images, voice and system data are cross-referenced against known problems.
SOLVE	Diagnostic reports, repair guides and maintenance schedules.



03 / 05

DESIGNED FOR GLOVED HANDS AND LOW TRUST

Most AI interfaces are designed for someone sitting down. This one is designed for someone standing next to a machine that is not working, holding a tool, and inclined to disbelieve the answer.

INPUT PARITY	Typing, speaking and photographing are equivalent ways to ask, not a text box with voice bolted on.
DARK AND LEGIBLE	High contrast, large type and big targets, for a plant room or a vehicle at night.
PROGRESSIVE TRUST	Name, email and password first. Identity on its own screen, with a progress bar.
PLAIN LANGUAGE	The FAQ answers in a technician's vocabulary, with no model names or token counts.

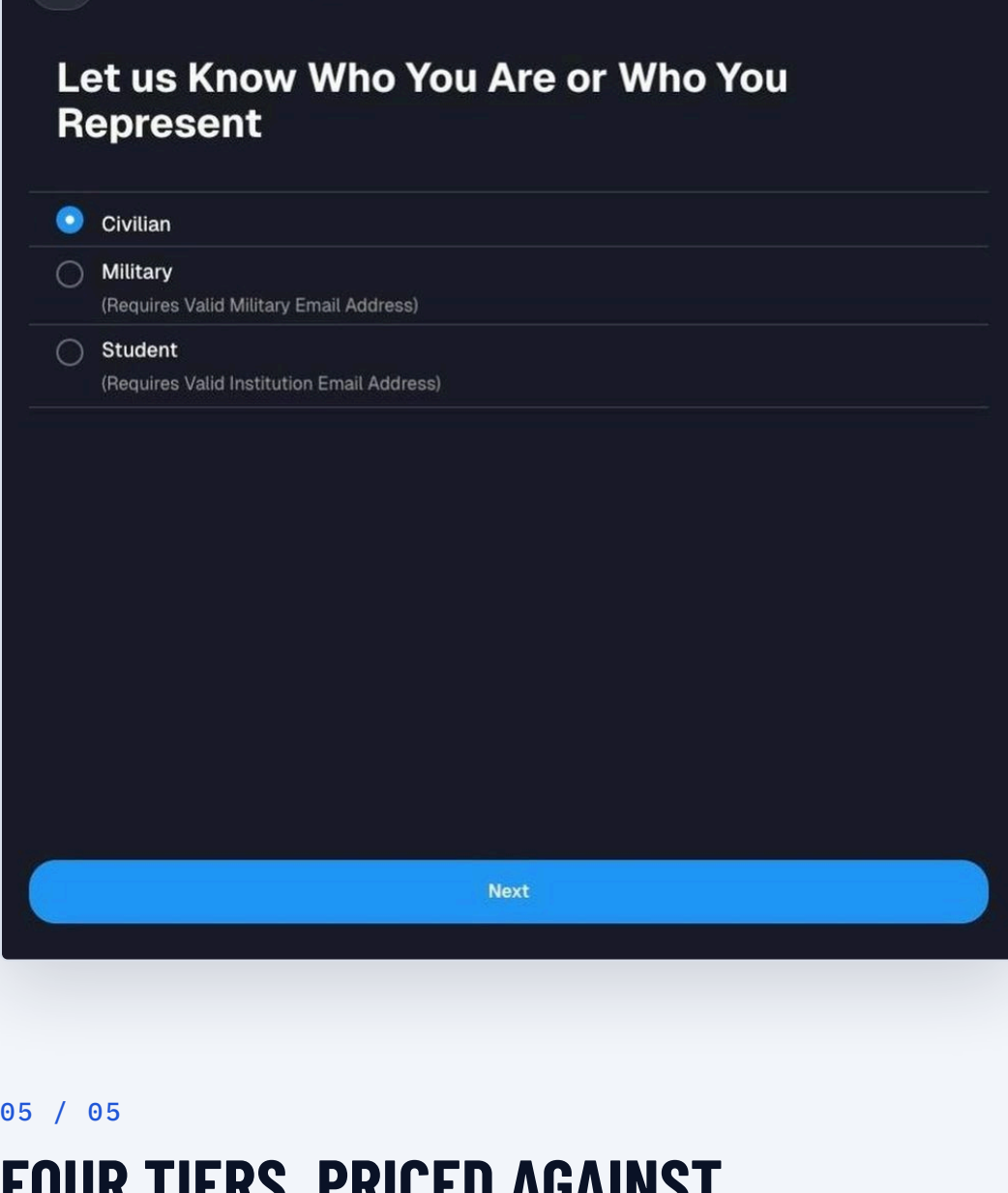
04 / 05

WHO YOU ARE DECIDES WHAT THE ASSISTANT CAN READ

The knowledge base is not one pool. Account type is chosen at sign-up, verified by email domain, and then decides what document repository the assistant may retrieve from. For a military account, retrieval never reaches into a shared corpus.

The boundary sits in the architecture, not in the terms of service.

Civilian · public manual database	Military · .mil verified, isolated knowledge base	Student · institution email
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WHAT EACH TIER SELLS	
FREE Proof the interaction works	\$0
PRO Capacity and modality	\$49/mo
MILITARY Isolation and auditability	\$149/mo
ENTERPRISE Control and integration	Custom

05 / 05

FOUR TIERS, PRICED AGAINST THREE DISTINCT BUYERS

The free tier proves the interaction. Pro sells capacity. The military tier sells isolation. Enterprise sells control. The step from Pro to Military is priced on where the documents live, which is what a defence buyer's security review asks about.

FREE · \$0	Public manual database, text chat, basic image analysis.
PRO · \$49/MO	Voice, unlimited image and video analysis, uploads and exportable transcripts.
MILITARY · \$149/MO	Everything in Pro, plus an isolated repository, encryption options and audit logs.
ENTERPRISE	On-premise option, API access, SSO and custom model training.

RESULTS & IMPACT

A WORKING PRODUCT, STATED AT THE LEVEL IT CAN BE DEFENDED

This is a capability engagement at the pre-measurement stage. The honest account of its impact is what now exists and can be demonstrated.

Diagnose	A working diagnostic path A technician reaches repair guidance from a photograph, a spoken question or a typed symptom, grounded in the equipment's own documentation.	Isolated	A defence-ready boundary Identity verification and an isolated knowledge base make the product addressable to a market a shared-corpus assistant cannot serve.
4 tiers	A monetisation structure Each tier maps to a distinct buyer and a distinct reason to upgrade.	Memory	Institutional memory Context-aware responses recall a unit's equipment history, so repeat conversations start further along.
Complete	A complete account lifecycle Sign-up, identity gating, code verification, Google sign-in, password recovery and plan selection are all built.	Live	A responsive web application Built on Next.js and hosted on Render as a responsive web application.

No user counts, revenue, diagnostic accuracy or repair-time figures are claimed. None were available to verify, and no analytics were in place at the time of review.

WHAT'S NEXT

THREE ADDITIONS THAT WOULD CHANGE THE PRODUCT

The architecture is retrieval-grounded. The next step is to make that visible to the technician, and to measure it.

VISIBLE SOURCE CITATIONS Document, revision and page beside every answer, so a technician can verify it once and trust it afterwards.	AN EQUIPMENT COVERAGE INDEX By manufacturer and equipment class. It answers "do you cover my fleet?" and lands model-specific searches.	INSTRUMENTATION Retrieval hit rate, time to answer, input-mode split and sign-up to first question conversion.
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QUESTIONS

ABOUT THIS AI MAINTENANCE ASSISTANT CASE STUDY

WHAT IS INTELLIMAINT AI?	+
HOW DOES THE ASSISTANT AVOID INVENTING ANSWERS?	+
CAN MILITARY TEAMS USE IT?	+

AN AI PRODUCT THAT NEEDS TO BE RIGHT, NOT JUST IMPRESSIVE?

Bytes & Partners builds retrieval-grounded AI products for domains where a wrong answer has a physical cost, and instruments the result so the impact can be proved rather than asserted.

START A CONVERSATION

DOWNLOAD THE FULL CASE STUDY